Electrical Power and Energy Systems 120 (2020) 105969

Contents lists available at ScienceDirect

Electrical Power and Energy Systems

journal homepage: www.elsevier.com/locate/ijepes

A new strategy for short-term ramp rate control in wind farms

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ARTICLE INFO

Keywords:
Wind power
Ramp control
Curtailment
Forecasts

ABSTRACT

Short-term drops in power output of wind farms may be a serious impediment to large-scale wind energy deployment in weak systems such as island grids. As opposed to operational measures countering this problem by generalized or *flat* curtailment, a novel approach based on short-term power forecasts is proposed in the present work. Analytical expressions for compliance with a given yet variable slope limit, achievable through Forecast-Based Curtailment (FBC), are derived for forecasts of varying accuracy. The cost-benefit relations of flat curtailment vs. FBC are compared. Output power change distributions used in this work are shown to be consistent with observational data from an operational wind farm. Numerical simulations based on both synthetic and operational wind farm data confirm the accuracy of the analytical expressions derived in this work.

1. Introduction

As the integration of renewable energy technologies into the electric grids around the world evolves, the level of variable generation increases, requiring new strategies to ensure a reliable service to the end customer [1]. Frequency regulation [2], i.e. maintaining the system frequency within a certain predefined range, is one of the basic tasks of the electric grid operator, achieved by balancing load and generation through a cascade of measures. When considering frequency regulation two important issues have to be dealt with: (1) the response of the system to changes in frequency, and (2) the mitigation of the variability of generation. Typically, large frequency drops occur in response to generator outages if the available short-time reserve margins are limited. However, with a more pervasive presence of variable generation in electric grids, output power fluctuations of wind farms come under increasing scrutiny, since they might strain reserve margins or even compromise frequency stability, potentially leading to the need for load shedding. Though the perception of system operators regarding the severity of wind farm power variability may occasionally differ from the empirical data the general point has to be conceded: wind farm power fluctuations have to be managed and potentially mitigated. Often, Energy Storage Systems (ESS) [3–6] are proposed to mitigate fluctuations in weak electric grids, such as island grids [4]. Though ESS are capable of damping fluctuations in an effective way, there may be no straightforward way of receiving remuneration for this service, or the corresponding income may be too low; the requirement of an ESS may then increase the cost of the wind farm far beyond the economically justifiable cost if integration of variable generation.

The grid integration of wind energy has been covered in a considerable body of research [5–7]. One line of investigation deals with the first point mentioned above, the response to variations in frequency, generally known as frequency support [2,8–10]. Responding to an *increase* in frequency is conceptually simple, since the wind farm output can be readily reduced by lowering the power extraction efficiency, e.g. by blade pitching. Upregulating, i.e. the response to a *decreasing* frequency, is more challenging since normally wind turbines are operated at the maximum power point for a given wind speed. Three schemes have been investigated in depth, both corresponding to the very short-term regulation normally referred to as primary regulation: (a) the extraction of kinetic energy from the rotor [1,8,9], also known as synthetic inertia [11,12], (b) *de-loading* [8,11] or de-rating by operating the wind turbine above the rated shaft speed for maximum power extraction, and de-loading by blade pitching [10]. The (modest) energy storage potential of the DC-link between rotor-side and grid-side converters in full-scale power converters has also been explored [10]. Synthetic energy schemes cause a strong reduction in rotor speed and must therefore carefully consider recovery after a disturbance. De-loading requires operating the wind turbine below nominal power, which is normally achieved by operating the rotor above rated speed; during a frequency drop, the rotor is decelerated and increases its output by moving towards higher efficiencies. Though initial research on de-loading focused on primary frequency regulation, it has also been used for the construction of ramping capabilities on longer time scales, such as the ones relevant for secondary regulation and market operations [13]. *Forecasting* wind power variability [14,15], on the other hand, can help improve grid operations and minimize reserve requirements. Several authors have stressed the importance of accurately predicting large ramps [16], both by meteorological [17] and stochastic modeling [18,19], and have developed corresponding solutions. Using battery storage systems [7,20] or combinations of battery systems and super-capacitors [21] the impact of wind power ramps can be mitigated, and potentially allow for a greater amount of wind power

https://doi.org/10.1016/j.ijepes.2020.105969
Received 7 October 2019; Received in revised form 4 January 2020; Accepted 24 February 2020
Available online 04 March 2020
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Nomenclature

α	Maximal rate of change of output power [MW/min] assumed by FBC
a	$= \alpha\Delta t$, where Δt=1 min
a_{sys}	Maximal change of output power [MW] tolerated by the system
δa	Maximal forecast error for x and y
$\hat{a}$	$= \lambda a$, normalized power change limit
β	Scale parameter of the Cauchy-Lorentz distribution
γ	Flat curtailment factor
$\mathbb{C}$	Compliance region
η	Mitigation efficiency
χ	$= x_{s,\mathrm{eff}}/P_0$, fluctuation scale, normalized to name-plate capacity
f_0	Nominal grid frequency
f_{XY}	Bivariate probability density function (pdf) for consecutive power changes
f_X, f_Y	Marginal pdfs
$f_{\delta X,\delta Y}$	Bivariate probability density function for forecast errors
Δf_1	Tolerable frequency deviation under normal operating conditions
Δf_2	Maximal frequency deviation before load shedding is activated
λ	Scale parameter of the Laplace distribution
ζ	Toggle variable for activating FBC
p, q	Parameters of the generalized Laplace distribution
δP	Curtailed power (power not delivered)
ρ	Compliance fraction
X	Random variable identifying power changes in first time interval
Y	Random variable identifying power changes in second time interval
$\delta x, \delta y$	Forecast errors for x and y
$\delta\hat{x}, \delta\hat{y}$	Normalized versions of δx and δy
x	A specific value of X
y	A specific value of Y
x_s	$= 1/\lambda$, characteristic scale [MW] of the Laplace distribution

capacity and reduced rates of *curtailment*, i.e. a generally involuntary reduction in output power mandated by the grid operator. A number of authors [22–24] have dealt with this subject. Yet another group of authors has dealt with the issue of the grid-wide minimization of wind power fluctuations [25,26], or the consideration of wind power ramps as a risk factor in unit commitment [27]. Empirical studies on wind power fluctuations, and in particular the distribution of ramps have also attracted the attention of a number of authors [28–30].

Curtailment of generation is often used as a measure of relief if an electric grid cannot cope of the amount of variable generation offered [31]. In many cases, the need for curtailment is a manifestation of insufficient transmission capacity in grids not originally designed for renewable energy integration. In other occasions, curtailment is used because of insufficient reserves to ensure frequency regulation or resource adequacy [4,31,32]. Grid operators or utilities may often set limits on the short-term rate of change, particularly in island grids. The Hawaiian utility HECO, e.g., requires wind farms at the Kahuku island grid to comply with an output power rate-of-change limit of 2 to 3 MW/min. Similarly, the Danish grid code (as discussed in [32]) under certain conditions imposes a 0.1 p.u. (per unit)/min ramp limit for both increasing and decreasing power. Rose & Apt [32] assessed the amount of energy not delivered under a number of scenarios, inspired in the Danish grid code, where compliance with the downward ramp limit was achieved by permanently de-loading or curtailing the wind farm. In the rest of the paper, we will refer to this kind of strategy as *flat curtailment*. Evidently, a high energy penalty has to be paid in this case, so smart approaches that avoid either additional upfront investment costs (such as battery storage systems) or losing part of the income because of energy not delivered are required. Such a strategy will be proposed in the present work.

This strategy allows a wind farm to reduce large negative power ramp rates, without having to recur to storage or permanent curtailment. It is driven by short-term forecasts and can be viewed as a self-regulating mechanism. The underlying idea, which will be referred to as *Forecast-Based Curtailment* (FBC), is quite intuitive but requires a comprehensive analytical framework in order to quantify the benefits and limitations of the method. Such a framework is laid out in the present work. In order to validate the analytical results, an algorithmic version of the FBC strategy using sets of synthetic data with appropriate distributions was implemented as well. The applicability of the proposed method to the real world is demonstrated by applying the FBC algorithm to power output time series of a real-operating wind farm and comparing the results to the ones obtained with the analytical theory.

This paper is structured as follows. In Section 2, first, the general problem statement is laid out, and some concepts and definitions will be provided. In Section 3, the analytical treatment of the novel method, i.e. FBC, is described in some detail. Closed expressions for the achievable benefits in terms of increased compliance with a given system reserve limit are derived first for the case of a perfect forecast, and are then extended to the general case of a finite forecast accuracy. Cost-benefit relations are derived by comparing the analytical expressions for ramp limit compliance vs. curtailment cost for both flat and

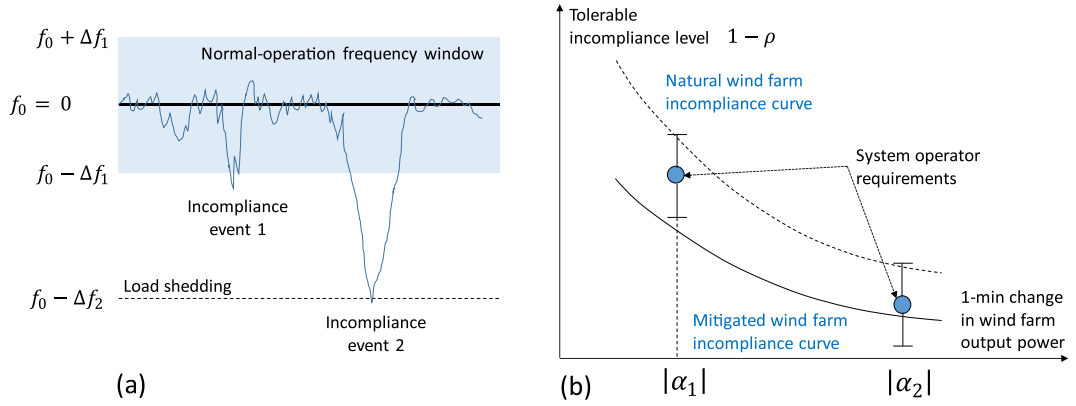


Fig. 1. Frequency stability and system operator requirements. (a) Illustration of two incompliance events. (b) Natural and unmitigated incompliance curves for a wind farm, combined with the system operator requirements.

forecast-based curtailment, demonstrating that FBC-compliance is a low-hanging fruit. Numerical simulations are used to validate the analytical formulas derived in Section 3. A study of how the FBC strategy performs with the observational data of an operating wind farm is studied in Section 4, where the assumptions and the analytical results of the FBC theory are validated by implementing the proposed curtailment strategy on observational wind farm data. Section 5 closes with a review of the results and an outlook into future work.

2. Basic concepts and definitions

2.1. Frequency stability and wind farm variability requirements

In order to illustrate the basic problem at hand Fig. 1 will be inspected first. Fig. 1(a) illustrates a typical situation in an island grid, where two levels of contingencies have been distinguished. Under normal-operation conditions the system frequency remains in the window $[f_0 - \Delta f_1, f_0 + \Delta f_1]$, where Δf_1 is defined by the applicable grid code. An incompliance event of the first type, with $f < f_0 - \Delta f_1$, may be caused by a sudden drop in wind farm power, leading to a mild contingency generally referred to as frequency violation. Grid codes are generally adamant about the value of Δf_1 but may be mute on the number of violations that can be tolerated during a given period of time. In any case, it can be assumed that the tolerable incompliance level $1 - \rho$ required from a wind farm for the corresponding rate of change in output power $\alpha = \mathrm{d}P/\mathrm{d}t < 0$ will be relatively high, as indicated in Fig. 1(b). For larger levels of $|\alpha|$, say, $|\alpha|_2$, corresponding to incompliance event 2 in Fig. 1(a), a load shedding procedure may have to be activated. Naturally, the tolerable incompliance levels can be expected to be lower in this case, although the actual figures may not be prescribed by the grid code and may be subject to discretional requirements by the grid operator; this has been indicated by the error bars in Fig. 1(b). Apart from the required (in)compliance levels (specified at discrete levels of power changes) a continuous curve $1 - \rho_0(a)$, indicating the incompliance level obtained by a normally operating wind farm for a given power change $a = \alpha\Delta t$ during the time interval Δt has been included in Fig. 1. $\rho_0(a)$ will be referred to as the *natural compliance curve*. In this paper, it will generally be assumed that $\Delta t = 1$ min, although the results are in principle applicable to other desired time scales. The natural compliance curve may or not allow to comply with the specific requirements enforced by the system operator, so some degree of mitigation may be required. Such a mitigation measure may be a BESS or Forecast-Based Curtailment (FBC), the strategy proposed in the present work. As shown in Fig. 1(b) the FBC-enhanced incompliance curve $1 - \rho_0(a) - \rho_{\mathrm{FBC}}(a)$ should be able to be lower than the tolerable incompliance level at the specified a locations.

2.2. Output power step change distributions

Let us now consider the net output of a given wind farm as a time series $t \to P(t)$ evaluated at discrete times t_n. Typically in this paper, one-minute intervals will be considered, so $t_{n+1} - t_n = \Delta t = 1\mathrm{min},\ \forall\, n$. Then, one-minute changes in output power for consecutive time intervals can be defined by

$$X = (P(t_n) - P(t_{n-1}))/\Delta t = (P_n - P_{n-1})/\Delta t, \tag{1a}$$

$$Y = (P(t_{n+1}) - P(t_n))/\Delta t = (P_{n+1} - P_n)/\Delta t, \tag{1b}$$

where the random variables X and Y have been introduced. As usual, capital letters are used to denote the random variables themselves, whereas lower-case variables (x and y in this case) are used for specific values assumed by the random variables.

Wind power changes on different time scales have been studied by a number of authors [33–36]. Wan [35] assessed power changes in the ERCOT system and found a fast decaying distribution for 1-min changes, though no pdf was proposed. Lee & Baldick [3] observed that a Laplace distribution represented an excellent fit to wind power changes at ERCOT at the minute and sub-minute limit level. In the present work a generalized Laplace (GL) distribution of the form

$$f_{X,\mathrm{GL}}(x) = \frac{1}{2(1+q)}(e^{-|\hat{x}|} + qpe^{-p|\hat{x}|}) \tag{2}$$

will be used, although some results will be shown only for the special case $q = 0$. $\hat{x} = \lambda(x - u_0)$ is a standardized variable, where $x_s = \lambda^{-1}$ is a characteristic scale and $p, q > 0$. The drift term u_0 will normally assumed to be zero but can be conveniently used to generalize the results to the case of a slowly moving mean; see Section 4.5. In addition to the distribution of Eq. (2) some results will also be derived for the heavy-tailed Cauchy-Lorentz (CL) distribution:

$$f_{X,\mathrm{CL}}(x) = \frac{1}{\pi\beta}\frac{1}{1 + (x - u_0)^2/\beta^2}. \tag{3}$$

An important assumption of this work (which will be revisited in Section 4) is that consecutive one-minute changes in wind farm output power are *uncorrelated*. This assumption is of course best checked against actual wind farm data but a theoretical rationale can also be built from some basic considerations regarding turbulence, the main driver for short-term fluctuations. If a van Karman turbulence spectrum is used (see e.g. [37])

$$\frac{fS_u(f)}{\sigma_u^2} = \frac{4fL_u/\bar{U}}{(1 + 70.8(fL_u/\bar{U})^2)^{5/6}} \tag{4}$$

the autocorrelation function can be calculated analytically to read

$$\kappa_u(\tau) = \frac{2\sigma_u^2}{\Gamma\left(\frac{1}{3}\right)}\left(\frac{\tau}{2T'}\right)^{1/3} K_{1/3}\left(\frac{\tau}{T'}\right), \tag{5}$$

where the characteristic time scale is related to the characteristic length scale according to $T' \cong 1.34L_u/\bar{U}$, and $K_{1/3}$ is the modified Bessel function of the second kind of order 1/3. Using $L_u = 73.5$ m for reference purposes, it is easily verified that $\kappa_u(\tau = 60\ \mathrm{s})/\sigma_u^2 = 10^{-3}$ for $\bar{U} = 10$ m/s and still $\kappa_u(\tau = 60\mathrm{s})/\sigma_u^2 \leq 3\%$ for an average wind speed of 5 m/s, illustrating the fact that the autocorrelation of turbulent wind speed variations decays very quickly and is usually negligible on the 1-min scale. Evidently, this simple argumentation does not consider correlations on longer time scales (10 min to h), which give rise to the well-known ramping characteristic of wind farms. It will be argued in Section 4 that the theoretical framework presented in this work can be extended to include ramps, though their effect on the net results of the present work will be generally negligible. For now, it will suffice to write the bi-variate probability density function (bi-pdf) of consecutive 1-min power changes as:

$$f_{XY}(x, y) = f_X(x) \times f_Y(y) \tag{6}$$

2.3. Natural compliance with the system slope limit

In the following a given limit for the maximum power drop during a 1-min interval which can be tolerated by the system will be assumed, i.e. $x_{\mathrm{lim}} \equiv -a$, $a > 0$. As discussed in Section 2.1, the specific interpretation depends on the system under study, but typically x_{lim} will correspond either to the largest negative variation of the wind farm output power which can be tolerated by the system, while still maintaining the frequency within the normal-operation frequency band, or the largest negative variation beyond which load has to be shed to preserve the stability of the system. The specific limits applicable to a given wind farm have to be determined from system impact studies conducted by the grid operator or consultants. For now, a will be assumed to be a parameter external to the wind farm.

The fraction of time during which the wind farm complies with this limit can now be stated in a straightforward way; in the case of the generalized Laplace model (Eq. (2)) the result is

$$\rho_{0,\mathrm{GL}} = \int_{-a}^{\infty} \mathrm{d}x f_X(x) = \int_{-\hat{a}}^{\infty} \mathrm{d}\hat{x} \hat{f}_{\hat{X}}(\hat{x}) = 1 - \frac{e^{\hat{a}} + q e^{-p\hat{a}}}{2(1+q)} \tag{7}$$

where $\hat{a} = \lambda(a + u_0)$ and $1/\lambda \equiv x_s$ [MW] is again a characteristic fluctuation scale. The natural compliance fraction for a Cauchy-Lorentz distribution (Eq. (2)) can also be calculated easily for completeness:

$$\rho_{0,\mathrm{CL}} = \left(\frac{1}{2} + \frac{1}{\pi}\arctan\hat{b}\right) \tag{8}$$

where $\hat{b} = (a + u_0)/\beta$ is defined in analogy with $\hat{a}$ as the scaled characteristic slope limit for the case of the CL distribution.

3. Analytical treatment and validations based on synthetic data

3.1. Forecast-based curtailment: the general idea

In this section, the main innovation of this work, the *Forecast-Based Curtailment* (FBC) strategy for short-term wind farm ramp control, and a number of analytical results will be obtained. The main output variables are the *FBC compliance fraction* $\rho_{\mathrm{FBC}}(\hat{a})$ and the *FBC curtailment cost* $\delta P_{\mathrm{FBC}}(\hat{a})$. It is also natural to define the *FBC efficiency*, i.e. the ratio of the FBC-mitigated incompliance events and the total number of incompliance occurrences. ρ_{FBC} can be compared to the compliance fraction which can be obtained with *flat curtailment*, which is referred to as a wind farm operating strategy where the output is de-rated by a constant factor with the respect to the maximum power which can be delivered at a given wind speed. Though a number of different conventional (i.e. not forecast-based) curtailment strategies can be imaged, flat curtailment is a useful reference against which FBC can be compared.

The case for FBC will be developed first based on the assumption of perfect forecasts, in order to make the point and determine an upper limit of the potential of the method. Real-world forecasts will of course have an associated uncertainty which has to be considered. Generalized expressions for the FBC compliance fraction and the curtailment cost for the case of a finite forecast uncertainty are also provided, allowing practical forecast approaches to be evaluated in terms of their usefulness for FBC. The prediction methods themselves are not subject of this paper and will be studied in follow-up work.

The basic idea of forecast-based curtailment (FBC) is to use the (presumably imperfect) knowledge of the power production values predicted to occur in the near future to adjust the power output *before* a significant drop or downward ramp occurs, thereby reducing the ramp to a level which can be handled by the regulation reserves available in the system. For an illustration of the idea of FBC the reader is referred to Fig. 2. *Natural* compliance occurs for $y \geq -a$; the additional compliance that can be achieved by implementing the FBC strategy corresponds to the triangular area in Fig. 2(b).

In (a) the power production of the wind farm is shown for three consecutive time steps (t_{n-1}, t_n, t_{n+1}), typically assumed to be 1-min averages, where $P_{n-1} \equiv P(t_{n-1})$ is the last value actually logged by the wind farm controller. P_n and P_{n+1} are predicted values available from a continuous short-term forecasting system. Possible options include nacelle-mounted LIDAR systems measuring the incoming wind, ground-based LIDAR [38] or SODAR systems placed at upstream locations, or tower-based anemometer measurements at conveniently located upstream locations. It will first be assumed that the predicted values P_n and P_{n+1} are accurate; further down this requirement will be relaxed in order to include finite forecast errors.

In Fig. 2(a) a slope limit incompliance situation has been illustrated for the interval $[t_n, t_{n+1}]$, where the change in power output is predicted to be more negative than a critical threshold ($y = P_{n+1} - P_n < -a < 0$, where $a = \alpha\Delta t > 0$ represents the maximum downward change in wind power output which can be tolerated by the system during the time interval Δt). It can be seen that – assuming a perfect knowledge of P_n and P_{n+1} – there is a finite margin for adjusting P_n as long as $\frac{1}{2}(x + y) \geq -a$. The FBC control algorithm is then required to perform the action

$$P_n \rightarrow P_n' = P_{n+1} + a \tag{9}$$

for all pairs of step changes $(x, y) \in \mathbb{C}_0$, where the set

$$\mathbb{C}_0(a) \stackrel{\mathrm{def}}{=} \left\{(x, y) \in \mathbb{R}^2 | y < -a \wedge \frac{1}{2}(x + y) \geq -a\right\} \tag{10}$$

will be referred to as the *ideal FBC-compliance region*. $\mathbb{C}_0$ represents the infinite half-space in the x-y plane contained between the lines $y = -2a - x$ and $y = -a$, as shown in Fig. 2(b). The additional compliance fraction that can be obtained by curtailing power at t_n (i.e. $P_n \rightarrow P_n'$) can be calculated by integrating $f_{XY}(x, y)$, the probability density for finding a power change x followed by a change y, over $\mathbb{C}_0$:

$$\rho_{\mathrm{FBC}}^{(0)}(a) = \int_{-a}^{\infty} \mathrm{d}x \int_{-x-2a}^{-a} \mathrm{d}y f_{XY}(x, y) \tag{11}$$

The superscript "0" identifies the ideal case of perfect predictions, which will be relaxed in Section 3.3 where finite forecast uncertainties will be considered. It is also convenient to define the *FBC efficiency* by

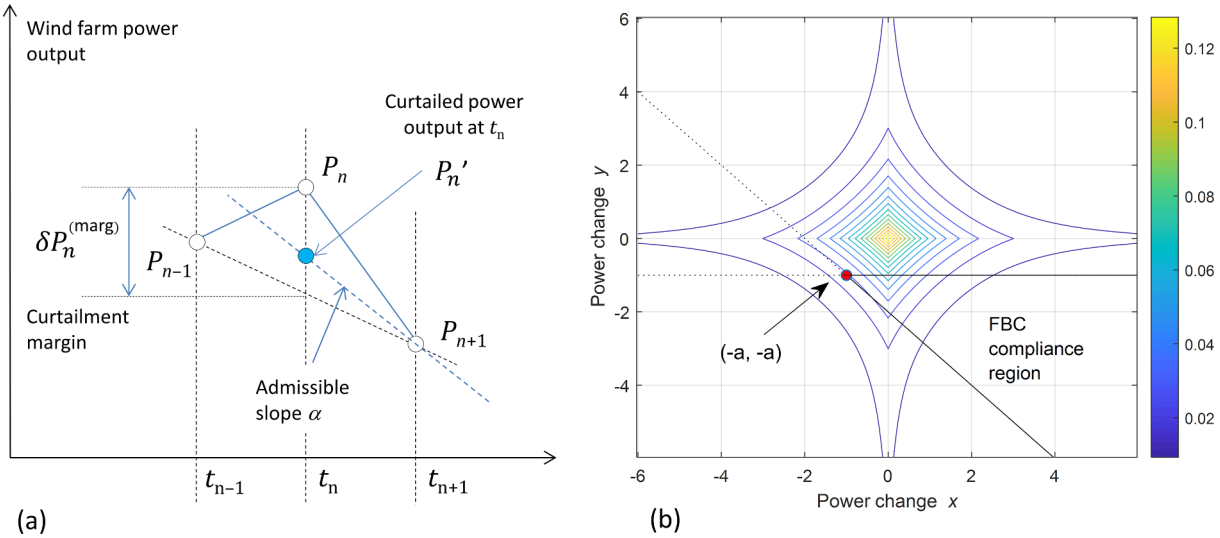


Fig. 2. Illustration of the concept of forecast-based curtailment for ramp control. (a) Power production at three consecutive time steps. (b) Indicative bivariate power change distribution and FBC compliance area for a given rate limit a.

$$\eta_{\mathrm{FBC}} \overset{\mathrm{def}}{=} \frac{\rho_{\mathrm{FBC}}}{1-\rho_0} = \frac{\rho_1 - \rho_0}{1-\rho_0}, \tag{12}$$

where $\rho_1 = \rho_0 + \rho_{\mathrm{FBC}}$ is the total compliance in the presence of FBC. Evidently, η_{FBC} measures the fraction of incompliance events (measured by $1-\rho_0$) which can be mitigated by FBC. For future reference we re-write Eq. (11) as

$$\rho^{(0)}_{\mathrm{FBC}}(\hat{a}) = \int_0^\infty \mathrm{d}u \int_{-u}^0 \mathrm{d}v f_{XY}(u-\hat{a}, v-\hat{a}) \equiv G(\hat{a}, \hat{a}), \tag{13}$$

where the function

$$G(c_1, c_2) \overset{\mathrm{def}}{=} \int_0^\infty \mathrm{d}u \int_{-u}^0 \mathrm{d}v f_{XY}(u-c_1, v-c_2) \tag{14}$$

has been introduced and $\hat{a}$ is a suitably normalized version of the system slope limit. Assuming again a factorizing bivariate probability density $f_{XY}(x, y)$ the expressions in Eq. (11) can be evaluated analytically. Assuming first a simple Laplace distribution (i.e. Eq. (2) with $q = 0$) and using normalized variables $\hat{x} = \lambda(x - u_0)$ and $\hat{y} = \lambda(y - u_0)$ i.e.

$$f_{XY}(x, y) = f_X(x) f_Y(y) = \frac{\lambda^2}{4} e^{-\lambda(|x|+|y|)} = \frac{1}{4} e^{-|\hat{x}|-|\hat{y}|}, \tag{15}$$

G can be evaluated for the Laplace case to read

$$G_{\mathrm{L}}(c_1, c_2) = \frac{1}{8} e^{-c_1-c_2}(-3 - 2c_1 + 4e^{c_1}), \tag{16}$$

leading to

$$\rho^{(0)}_{\mathrm{FBC,L}} = \frac{1}{8} e^{-2\hat{a}}(-3 - 2c_1 + 4e^{\hat{a}}). \tag{17}$$

Consequently, the FBC efficiency can be calculated to read

$$\eta^{(0)}_{\mathrm{FBC,L}} = 1 - \left(\frac{3}{4} + \frac{\hat{a}}{2}\right) e^{-\hat{a}} \tag{18}$$

In analogy with Eq. (11), the subscript "0" refers to perfect forecasts. It is interesting to observe that $\eta^{(0)}_{\mathrm{FBC,L}} \to 1$ for $\hat{a} \to \infty$, i.e. the ability of FBC to mitigate incompliance events becomes perfect in the limit of large tolerable system slopes. Though a high mitigation efficiency may seem irrelevant when natural compliance is already high, it should be remembered from the discussion in Section 2.1 that (1) the relevant system limit (such as normal-operation frequency violations) is typically large compared to the scale of the wind farm fluctuations (i.e. $\hat{a} \gg 1$) and (2) the total number of tolerable incompliance events may be quite small, so a high mitigation efficiency is critically important even if the natural compliance rate is already very high. It is also noted that the FBC efficiency is low for small $\hat{a}$ values ($\eta^{(0)}_{\mathrm{FBC,L}} \to 1/4$ for $\hat{a} \to 0$) which is both unsurprising and irrelevant for practical applications.

Eqs. (11) and (12) can also be evaluated for the more general case of a generalized Laplace (GL) distribution (Eq. (2) with $q > 0$), leading to the somewhat more complex expressions below:

$$\rho^{(0)}_{\mathrm{FBC,GL}}(\hat{a}) = \frac{e^{-\hat{a}} + q e^{-p\hat{a}}}{2(1+q)} + A_{p,q} e^{-2\hat{a}} + B_{p,q} q e^{-2p\hat{a}} + C_{p,q} e^{-\hat{a}(1+p)} \tag{19}$$

$$A_{p,q}(\hat{a}) = -\frac{1}{8}\left((3 + 2\hat{a}) - \frac{4p^2}{1-p^2}\frac{q}{(1+q)^2}\right) \tag{19a}$$

$$B_{p,q}(\hat{a}) = -\frac{1}{8(1+q)^2}\left((3 + 2\hat{a}p)q + \frac{4}{1-p^2}\right) \tag{19b}$$

$$C_{p,q} = -\frac{q}{4(1+q)^2} \tag{19c}$$

The first term in Eq. (19) can be recognized as $1 - \rho_{0,\mathrm{GL}}$ (see Eq. (7)); consequently, the FBC efficiency can be seen asymptotically tend towards 100% for large tolerable slope values $\hat{a}$, given that all other terms in Eq. (19) tend to zero for $\hat{a} \to \infty$.

When comparing the expression for the simple and the generalized Laplace case a question naturally arises: what should be *effective scale* to be used in the simple Laplace model, given that the generalized Laplace model involves two scales (x_s and x_s/p)? In Section 4, when dealing with observational data, it will be shown that a weighted average of both scales is an appropriate choice:

$$x_{s,\mathrm{eff}} = x_s \frac{1 + q/p}{1+q} \tag{20}$$

In Section 4 it will also be argued that a Cauchy-Lorentz (CL) probability density is usually not a good fit to power change distributions of operating wind farms, but for completeness the corresponding results will be reported for CL as well. It can be shown readily that

$$\rho^{(0)}_{\mathrm{FBC,CL}}(\hat{b}) = \frac{1}{8} - \frac{1}{2\pi^2}\arctan^2 \hat{b} \tag{21}$$

where $\hat{b} = (a + u_0)/\beta$ is the normalized version of the system slope limit as before. Using Eq. (8) the FBC efficiency for the Cauchy-Lorentz case can be evaluated to read

$$\eta^{(0)}_{\mathrm{FBC,CL}} = \frac{1}{4}\left(1 + \frac{2}{\pi}\arctan \hat{b}\right). \tag{22}$$

Unlike its Laplace counterpart, the CL FBC efficiency asymptotically approaches ½ for large normalized $\hat{b}$ values, so the mitigation capabilities of Forecast-Based Curtailment are limited to a maximum of 50% of all incompliance events. Fortunately, wind farm power output changes do not seem to be CL-distributed.

3.2. FBC as a tool to increase the size of the wind farm

In order to put the results obtained so far in some context it is useful to think of Forecast-Based Curtailment as a means to increase the installed capacity of a wind farm in an electric grid constrained by frequency stability. To do so, it will be recalled that the system operator requirement can be written as $\rho(a_{\mathrm{crit}}/x_s) = \rho_{\mathrm{req}}$, where a_{crit} is the critical slope and ρ_{req} is the required compliance level at a_{crit} enforced by the system operator; x_s the characteristic scale of the fluctuations and ρ the total compliance level (natural + FBC). Assuming an *effective Laplace distribution* with a fluctuation scale $x_{s,\mathrm{eff}}$ for simplicity, a back-of-the-envelope estimation can then be made for the permissible wind power plant capacity. Considering first the natural compliance case an expression for the acceptable wind farm size can be obtained by solving $\rho_0(a_{\mathrm{crit}}/x_{s,\mathrm{eff}}) = \rho_{\mathrm{req}}$ for $x_{s,\mathrm{eff}}$ and writing the installed power capacity as $P_0 = x_{s,\mathrm{eff}}/\chi$, where χ is a site-specific constant:

$$P_{0,\max} = \frac{x_{s,\mathrm{eff}}}{\chi} = -\frac{a_{\mathrm{crit}}}{\chi \ln[2(1-\rho_{\mathrm{req}})]}. \tag{23}$$

Similarly, the *FBC-enhanced wind farm size* can be calculated by solving the following equation for x_s and using again $P_0 = x_{s,\mathrm{eff}}/\chi$ to estimate the installable wind farm capacity:

$$\rho_0(a_{\mathrm{crit}}/x_{s,\mathrm{eff}}) + \rho^{\mathrm{FBC}}(a_{\mathrm{crit}}/x_{s,\mathrm{eff}}) = \rho_{\mathrm{req}} \tag{24}$$

The more general case of a generalized Laplace distribution (i.e. Eq. (2) with $q > 0$) can be solved in a completely analogous way by first determining the scale x_s of the first exponential term and then applying Eq. (21) to determine the effective scale. A practical calculation based on the observational data from an observational wind farm will be examined in Section 4. It should be noted that the considerations of this subsection are based on the argument of a constant $x_{s,\mathrm{eff}}/P_0(\equiv\chi)$ ratio. This is slightly conservative, since larger wind farms tend to have smaller relative variations. (See, e. g., Wan et al. [35] for comprehensive discussion of empirical results obtained from operating wind farms). The actual benefits from FBC in terms of an increased wind farm tolerable with the system ramp rate limit would therefore be somewhat larger.

3.3. FBC vs. flat curtailment and FBC-enhanced compliance

From the results obtained in Section 3.1 it should have become clear that potential of FBC for mitigating incompliance is large. However, a few basic questions arise, such as: (1) how does the performance of FBC compare with that of flat curtailment, (2) can flat and forecast-based curtailment be combined to achieve a given level of compliance, and (3) what is the impact of FBC on the power not delivered, particularly compared to flat curtailment?

In order not to distract the discussion from answering these basic questions only Laplace pdfs (i.e. Eq. (2) with $q = 0$) will be considered in this subsection and the rest of Section 3; generalized Laplace distributions will be revisited in Section 4. It turns out that the answers to questions (1) and (2) can be lumped into one by defining the *FBC-enhanced compliance*, defined as the total compliance in the presence of either flat or forecast-based curtailment, or a combination of both:

$$\rho(\hat{a},\gamma,\zeta)=\rho_{0,\mathrm{L}}+\rho^{(0)}_{\mathrm{FBC,L}}=1-\frac{1}{2}e^{-\hat{a}/\gamma}+\frac{\zeta}{8}e^{-2\hat{a}/\gamma}\left(-3-2\frac{\hat{a}}{\gamma}+4e^{\hat{a}/\gamma}\right) \tag{25}$$

As before, γ is the flat curtailment level (with $\gamma = 1$ corresponding to no curtailment), and ζ is a binary variable corresponding to activated ($\zeta = 1$) or idle ($\zeta = 0$) FBC. Note that ρ only depends on $\hat{a}/\gamma$. The benefits of forecast-based curtailment are illustrated in Fig. 3 where combinations of natural compliance, flat curtailment, and FBC have been displayed; the inset of the figure has the contributions from FBC alone. It can be seen that FBC adds of up to 1/8 or 12.5 percent points to compliance (for $\hat{a} \to 0$), and decreases to a few percent points for larger $\hat{a}$. It is important to observe in Fig. 3 that a flat curtailment to 55% power output is required to produce a similar effect on total compliance as FBC (with FBC performing even better at small $\hat{a}$ values), a quite remarkable finding given the very low curtailment cost for FBC, explained in detail below.

Depending on the tolerable slope and the required compliance level, FBC alone can lead to compliance, with no other technical fixes (say, battery energy storage systems or fast-starting Diesel engines) required. If the system requirements are very severe, or the size of the wind farm is relatively large compared to the system size, then flat curtailment may be used in addition to FBC. Though flat curtailment comes at a substantial energy cost, as shown above, some of the cost can be mitigated by operating in tandem with FBC, since flat curtailment increases the effectiveness of FBC by downscaling the tolerable slope to $\hat{a}/\gamma$; see inset of Fig. 3.

3.3.1. The cost of compliance

The energy cost of a forecast-based curtailment can be assessed by observing that an individual curtailment event leads to a reduction in power at time t_n of $P_n^{'} - P_n = (P_n^{'} - P_{n+1}) + (P_{n+1} - P_n) = (a + y)\Delta t$, as long as (x, y) is in the compliance region (i.e. $(x, y) \in \mathbb{C}_0$). Taking the compliance cost as a positive number we have:

$$P_{\mathrm{FBC,curt}} \overset{\mathrm{def}}{=} \begin{cases} |P_n^{'} - P_n| = -(y + a)\Delta t & (x, y) \in \mathbb{C}_0 \\ 0 & (x, y) \notin \mathbb{C}_0 \end{cases} \tag{26}$$

The expected value of the compliance cost can then be calculated as

$$\delta P^{\mathrm{FBC},0} \equiv P_{\mathrm{FBC,\,curt}} = -\frac{\Delta t}{\alpha}\int_{-\hat{a}}^{\infty} \mathrm{d}\hat{x} \int_{-\hat{x}-2\hat{a}}^{-\hat{a}} \mathrm{d}\hat{y}\,(\hat{y} + \hat{a}) f_{XY}(\hat{x}, \hat{y}) \tag{27}$$

In analogy to Eq. (9) the superscript "0" refers to perfect forecasts. Eq. (13) can be conveniently re-written as

$$\delta P^{\mathrm{FBC},0} = -\frac{\Delta t}{\alpha}\int_0^{\infty} \mathrm{d}u \int_{-u}^{0} \mathrm{d}v v f_{XY}(u - \hat{a}, v - \hat{a}) \equiv \chi P_0 \times H(\hat{a}, \hat{a}), \tag{28}$$

where the name-plate wind farm capacity P_0 and the function

$$H(c_1, c_2) \overset{\mathrm{def}}{=} \int_0^{\infty} \mathrm{d}u \int_{-u}^{0} \mathrm{d}v \;\; v \;\; f_{XY}(u - c_1, v - c_2) \tag{29}$$

have been introduced for future reference. χ is the ratio between the 1-min fluctuation scale (in MW) and P_0 (also in MW) and is typically of the order of a few percent. For the case of the bi-variate distribution of Eq. (8) and assuming $q = 0$ again, H can be evaluated to read:

$$H(c_1, c_2) = \frac{1}{16}e^{-c_1-c_2}(-7 - 6c_1 - 2c_1^2 + 8e^{c_1}) \tag{30}$$

In analogy to the compliance fractions a total curtailment cost can

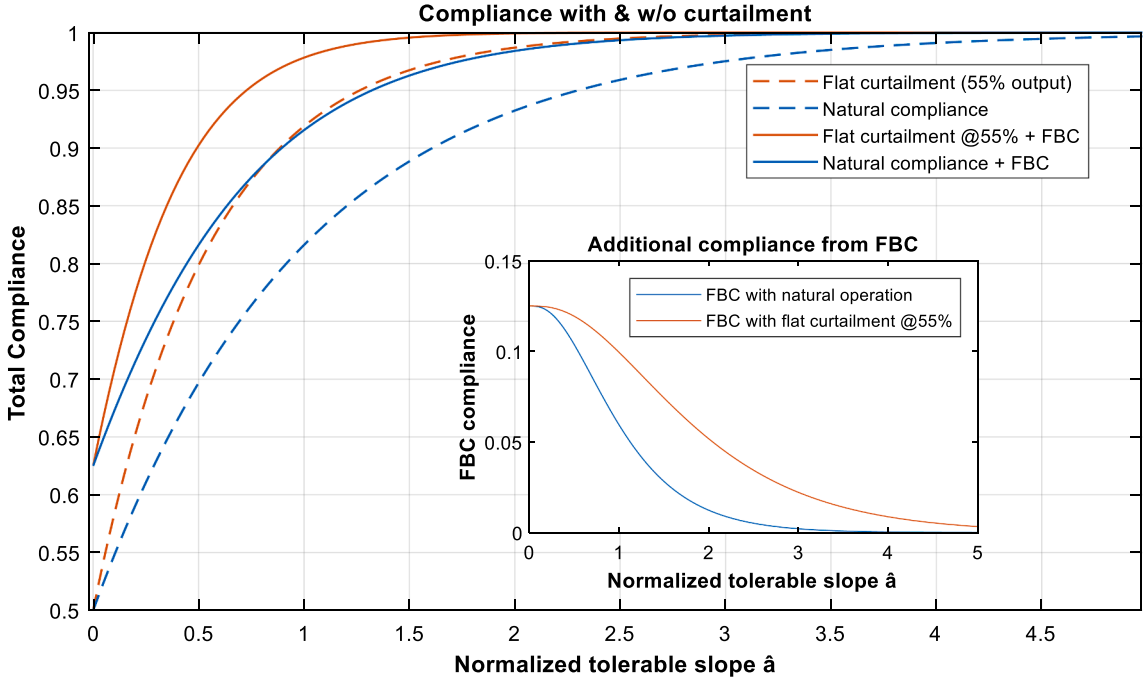


Fig. 3. Analytical results – ideal case: compliance achievable for a given tolerable ramp rate with combinations of natural compliance, flat curtailment, and forecast-based curtailment (FBC). *Large figure*: total compliance. *Inset*: contributions from FBC alone.

be defined as the total amount of power curtailed through either flat curtailment (δP^{flat}), FBC ($\delta P^{\text{FBC},0}$), or a combination of both:

$$\delta P^{\text{tot}} = \delta P^{\text{flat}} + \delta P^{\text{FBC},0} = P_0\left\{(1-\gamma) + \frac{\zeta}{16}e^{-2\hat{a}/\gamma}\left(-7 - 6\frac{\hat{a}}{\gamma} - 2\left(\frac{\hat{a}}{\gamma}\right)^2 + 8e^{\hat{a}/\gamma}\right)\right\} \quad (31)$$

It is now possible to compare the cost-benefit trajectories for flat and forecast-based curtailment, respectively. In Fig. 4(a) the flat compliance $\rho_0 = 1 - \frac{1}{2}e^{-\hat{a}/\gamma}$ has been plotted against the relative compliance cost $\delta P^{\text{flat}}/P_0 = 1-\gamma$, illustrating the high energy cost associated with flat curtailment, as demonstrated by the low slope of the ρ_0 vs$1-\gamma$ curve. Forecast-based curtailment (with $\gamma = 1$), on the contrary, has a very low energy penalty, as shown in Fig. 4(b), where the parametric curve $(\delta P^{\text{FBC},0}(\hat{a}), \rho^{\text{FBC},0}(\hat{a}))$ has been plotted.

Taking the case $\hat{a} = 1$ as an example, it can be seen that compliance can be increased by 10 percent points at a curtailment cost (i.e. power not delivered) of only about 0.55%. Evidently, in this quite challenging case (where the tolerable slope a is equal to the characteristic scale of fluctuations α^{-1}) the no-curtailment compliance is low to start with (around 0.82, see Fig. 4(a)), so in this case perfect compliance cannot be achieved with FBC alone. Is worth pointing out, however, the abysmal difference in curtailment cost for the compliance increase from 82% to 92% obtained for the $\hat{a} = 1$ case, since an energy loss of about 42% would result if flat curtailment were used instead of FBC, as evidenced by Fig. 4(a). As can be seen from Fig. 4 in the more realistic case of $\hat{a} \geq 3$ complete or near complete compliance can be achieved with FBC alone, at an almost negligible energy cost.

The analytical results presented so far can be validated by numerical simulations using synthetic power change time series as the input. An algorithmic version of the FBC strategy described in Section 3.1 was implemented for that purpose. Fig. 5 illustrates some of the results obtained with forecast-based curtailment alone (no flat curtailment, $\gamma = 1$) applied to a simulated time series with a Laplace distribution ($q = 0$) and assuming perfect forecasts. Fig. 5(a) highlights all events (circles) within a 2000-min stretch that would have incurred incompliance with the preset slope limit of $\hat{a} = 3$ assumed in this case. It can be seen that even in this quite challenging case (with the maximum tolerable slope being only 3 times the characteristic scale of the fluctuations) almost all of the incompliance events can be successfully mitigated (filled circles). A more representative view is presented in Fig. 5(b) where the probability density function (pdf), calculated from a full-year equivalent of time steps, has been plotted both for the unmitigated and the FBC-mitigated case. It can be seen that the mitigated pdf settles at a roughly constant value of 5×10^{-4} for step changes $x < -3$, far below the tail of the original distribution.

The quantitative predictions of the analytical theory are tested, again for the case of perfect forecasts, in Fig. 6(a) and (b), where the simulated compliance level for natural compliance ($\gamma = 1$) and FBC-enhanced natural compliance have been plotted, as well as the cost-benefit trajectory for the FBC-enhanced natural compliance case; a full-year equivalent simulation window has been used again. It can be seen that the analytical results are practically indistinguishable from the simulation results.

3.4. Compliance with non-ideal forecasts

So far, it has been assumed that the forecast values P_n and P_{n+1} are accurate. In practice, of course, there will be uncertainties associated with P_n and P_{n+1}, or equivalently, the forecast step changes X and Y. Then, x and y have to be replaced by $\hat{x} + \delta\hat{x}$ and $\hat{y} + \delta\hat{y}$, where $\delta\hat{x}$, $\delta\hat{y}$ ($\in \mathbb{R}$) are values of the random variables δX, δY with a suitable probability density function (pdf). Assuming $\hat{x}$ and $\hat{y}$ are the true step changes, then $\hat{x} + \delta\hat{x}$ and $\hat{y} + \delta\hat{y}$ are the changes predicted by the forecast algorithm. The latter, unaware of the forecast error, will perform the action

$$\hat{y}' \equiv \hat{y} + \delta\hat{y} \xrightarrow[\text{set to}]{} \hat{y}' = -\hat{a} \quad (32)$$

$$\text{if}(\hat{x}', \hat{y}') \equiv (\hat{x} + \delta\hat{x}, \hat{y} + \delta\hat{y}) \in \mathbb{C}_0(\hat{a}), \quad (33)$$

allowing for the possibility that the adjusted true slope of the second time interval $[t_n, t_{n+1}]$, i.e. $-\hat{a} - \delta\hat{y}$, will be more negative than $-\hat{a}$ (if $\delta\hat{y} > 0$). Compliance can, however, still be achieved if we now distinguish between the true *system slope limit* $\hat{a}_{\text{sys}} \overset{\text{def}}{=} \hat{a} + \delta\hat{a}$ and the *operational slope limit* $\hat{a}$, where $\delta\hat{a} > 0$ is the maximal forecast uncertainty for y, i.e. $\delta\hat{y} \in [-\delta\hat{a}, \delta\hat{a}]$. It is then readily verified that the adjusted true slope $-\hat{a} - \delta\hat{y} \geq -\hat{a} - \delta\hat{a} = -(\hat{a} + \delta\hat{a}) = -\hat{a}_{\text{sys}}$, i.e. compliance is ensured. If we assume for simplicity that the forecast uncertainty for the first time interval $[t_{n-1}, t_n]$ is the same as for the second one, i.e. $\delta\hat{x} \in [-\delta\hat{a}, \delta\hat{a}]$, then it can be readily verified that the true average slope of the interval $[t_{n-1}, t_{n+1}]$ is less negative than the system limit, i.e. $\frac{1}{2}(\hat{x} + \hat{y}) \geq -\hat{a} - \frac{1}{2}(\delta\hat{x} + \delta\hat{y}) \geq -\hat{a} - \delta\hat{a} = -(\hat{a} + \delta\hat{a}) = -\hat{a}_{\text{sys}}$, so

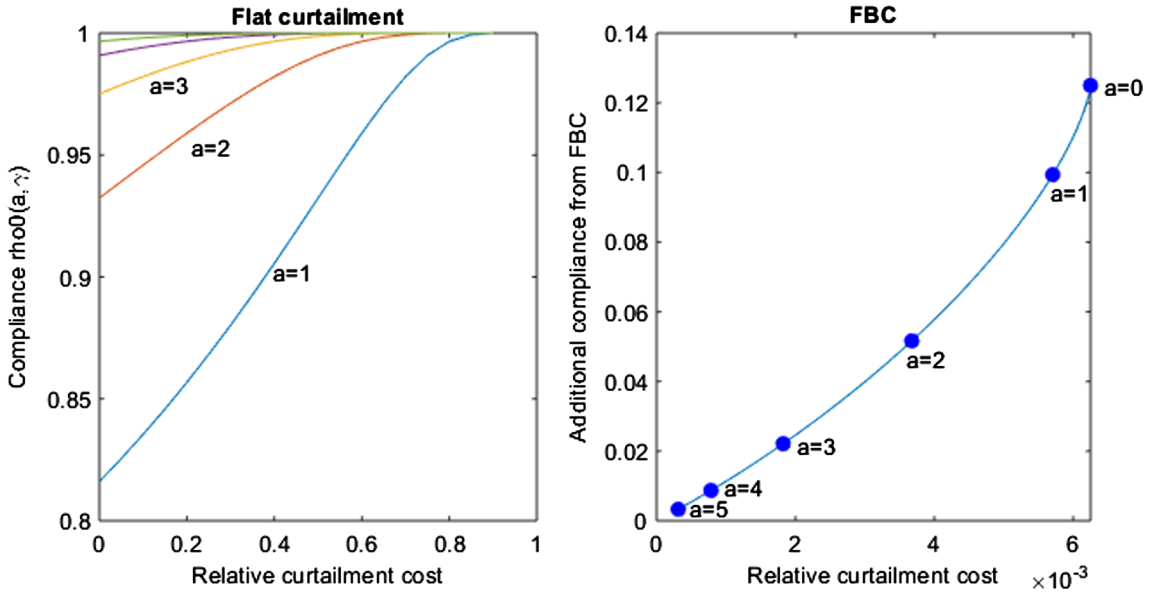


Fig. 4. Analytical results – ideal case: compliance vs curtailment trajectories for flat curtailment ($\zeta = 0$) (left graph), and forecast-based curtailment ($\zeta = 1$, $\gamma = 1$, $\chi = 0.1$), right graph.

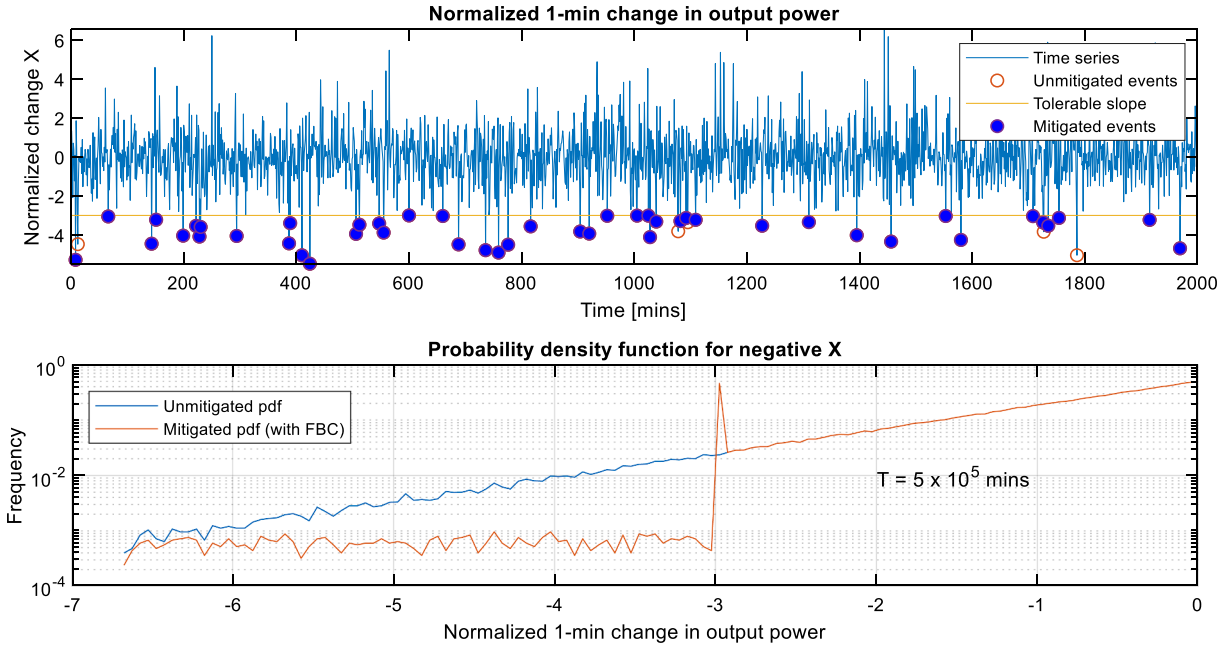


Fig. 5. Simulation results – ideal case. *Upper graph*: part of the time series simulated for one year, showing incompliance events and the tolerable slope ($\hat{a} = 3$ in this case). Filled circles indicate compliance after FBC, open circles correspond to remaining incompliance after FBC. *Lower graph*: Probability density function for negative step changes, both for the original case and after FBC.

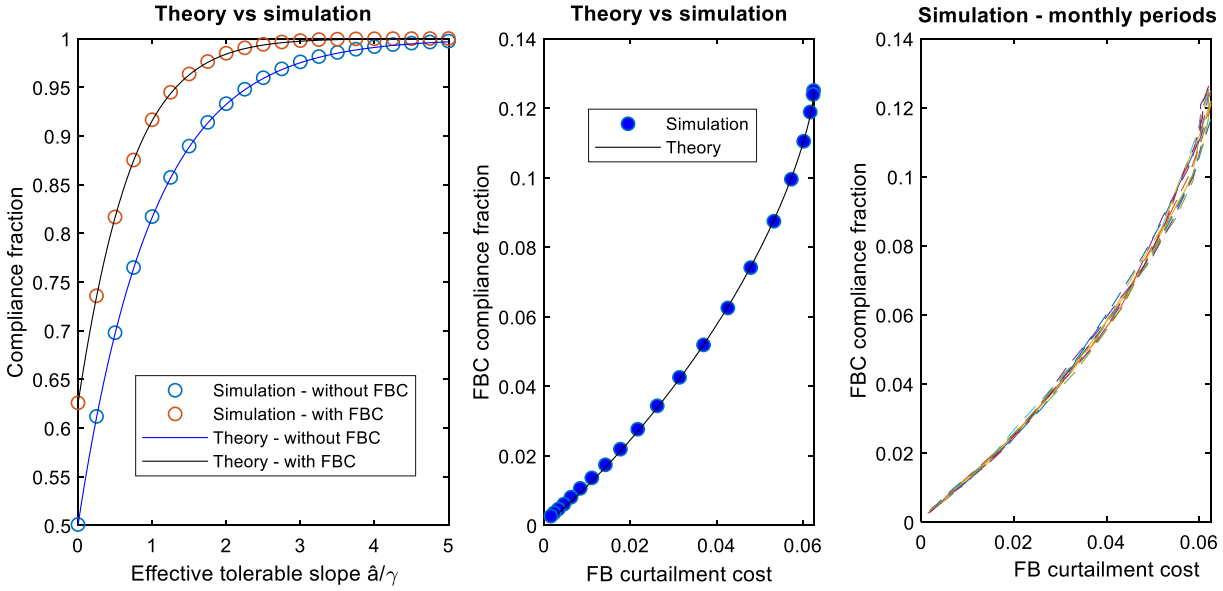


Fig. 6. Simulation vs analytical results – ideal case. Left graph: natural vs FBC-enhanced compliance for the case of no flat curtailment ($\gamma = 1$). Middle graph: FBC-compliance vs. compliance cost. *Dots*: simulation, *continuous line*: analytical results. Right graph: FBC-compliance vs. compliance cost for monthly periods.

sufficient curtailment margin will always be available as long as $(\hat{x} + \delta\hat{x}, \hat{y} + \delta\hat{y}) \in \mathbb{C}_0(\hat{a}) = \mathbb{C}_0(\hat{a}_{\text{sys}} - \delta\hat{a})$. In other words, the FBC procedure continues to ensure compliance in the presence of forecast uncertainties, as long as $\hat{a}$ is interpreted as the operational limitand distinguished from the true system limit $\hat{a}_{\text{sys}} = \hat{a} + \delta\hat{a}$.

Condition (33) can be expressed more conveniently in terms of the true step changes$x$ and y:

$$(\hat{x}, \hat{y}) \in \mathbb{C}_{(\delta\hat{x},\delta\hat{y})} = \left\{(\hat{x}, \hat{y}) \in \mathbb{R}^2 \middle| \hat{y} < -\hat{a} - \delta\hat{y} \wedge \frac{1}{2}(\hat{x} + \hat{y}) \geq -\hat{a} - \frac{1}{2}(\delta\hat{x} + \delta\hat{y})\right\} \tag{34}$$

Though this procedure does indeed ensure compliance in the presence of uncertainty, it should be noted that it does so at the expense of introducing (forecast-based) curtailment for some events whose slope $\hat{y}$ is less negative than the permitted system limit, i.e. cases with *natural compliance*. It is therefore necessary to introduce the concept of the *net compliance gain*, which identifies the compliance obtained *in excess of natural compliance*. For a given set of uncertainty values *the net compliance region* is given by

$$\mathbb{C}^{\text{net}}_{(\delta u,\delta v)} = \left\{(\hat{x}, \hat{y}) \in \mathbb{R}^2 | y < -\hat{a}_{\text{sys}} \wedge \frac{1}{2}(\hat{x} + \hat{y}) \geq -\hat{a}_{\text{sys}} + \frac{1}{2}(\delta u + \delta v)\right\}, \tag{35}$$

where the quantities $\delta u \equiv \delta\hat{a} - \delta\hat{x}$ and $\delta v \equiv \delta\hat{a} - \delta\hat{y}$ have been introduced. Note that δu, $\delta v \in [0, 2\delta\hat{a}]$. The net compliance gain $\rho^{\text{net}}_{\text{FBC}}$ for a given pair of forecast uncertainties $(\delta u, \delta v)$ can then be readily formulated as

$$\rho^{\text{net}}_{\text{FBC}}(\delta u,\,\delta v) = \int_{-\hat{a}_{\text{sys}}+\delta u+\delta v}^{\infty} \mathrm{d}\hat{x} \int_{-x-2\hat{a}_{\text{sys}}+\delta u+\delta v}^{-\hat{a}_{\text{sys}}} \mathrm{d}\hat{y} f_{XY}(\hat{x},\,\hat{y}), \tag{36}$$

or more conveniently as:

$$\rho^{\text{net}}_{\text{FBC}}(\hat{a}_{\text{sys}},\,\delta u,\,\delta v) = G(\hat{a}_{\text{sys}}-\delta u-\delta v,\,\hat{a}_{\text{sys}}), \tag{37}$$

where $G(c_1, c_2)$ is the function defined in Eq. (10). With this expression the expected value of the net compliance gain $\rho^{\text{net}}_{\text{FBC}}(\hat{a}_s,\,\delta u,\,\delta v)$ obtained with FBC can be formulated as

$$\langle \rho^{\text{net}}_{\text{FBC}} \rangle = \int_0^{2\delta\hat{a}} \mathrm{d}(\delta u) \int_0^{2\delta\hat{a}} \mathrm{d}(\delta v) \rho^{\text{FBC,net}}(\hat{a}_s,\,\delta u,\,\delta v) f_{\delta U,\delta V}(\delta u,\,\delta v), \tag{38}$$

where $f_{\delta U,\delta V}(\delta u,\,\delta v)$ is the (bi-variate) probability density function of the forecast errors. It should be remembered that $\delta u = \delta\hat{a} - \delta\hat{x} \in [0,\,2\delta\hat{a}]$ and $\delta v = \delta\hat{a} - \delta\hat{y} \in [0,\,2\delta\hat{a}]$. Assuming uncorrelated forecast errors with a uniform probability density in the interval $[0,\,2\delta\hat{a}]$ for simplicity, expression (36) can be evaluated using Eq. (35). After some algebra, the following expressions are obtained; $q = 0$ as before:

$$\langle \rho^{\text{net}}_{\text{FBC}} \rangle = -\frac{1}{8}(7+2\hat{a}_{\text{sys}})C(\delta\hat{a})e^{-2\hat{a}_{\text{sys}}+2\delta\hat{a}} + \frac{S(\delta\hat{a})}{2}e^{-2\hat{a}_{\text{sys}}+3\delta\hat{a}} + \frac{1}{2}e^{-\hat{a}_{\text{sys}}} \tag{39}$$

where

$$C(\delta\hat{a}) = \frac{\cosh(2\delta\hat{a})-1}{2(\delta\hat{a})^2} \tag{40}$$

$$S(\delta\hat{a}) = \frac{\sinh(\delta\hat{a})}{\delta\hat{a}} \tag{41}$$

In the limit of vanishing forecast uncertainties ($\delta\hat{a} \to 0$) the ideal-case Eq. (11) with $\hat{a} = \hat{a}_{\text{sys}}$ is recovered as$C,\, S \to 1$.

3.4.1. Curtailment cost with uncertainties

In order to calculate the curtailment cost in the presence of uncertainties, first the *curtailment region* can be stated:

$$\mathbb{C}^{\text{curt}}_{(\delta u,\delta v)} = \left\{(\hat{x},\,\hat{y}) \in \mathbb{R}^2 | y < -\hat{a}_{\text{sys}} + \delta v \wedge \frac{1}{2}(\hat{x}+\hat{y}) \geq -\hat{a}_{\text{sys}} + \frac{1}{2}(\delta u+\delta v)\right\}. \tag{42}$$

In analogy to Eq. (12) the power cost of an individual curtailment event in the presence of forecast uncertainty can now be written as

$$P^{\text{curt}}_{\delta v} \overset{\text{def}}{=} \begin{cases} |P_n' - P_n| = -(\hat{y} + \hat{a}_{\text{sys}} - \delta v)\Delta t & (x,\,y) \in \mathbb{C}_0 \\ 0 & (x,\,y) \notin \mathbb{C}_0 \end{cases} \tag{43}$$

The average value of the curtailed power in the presence of a fixed pair of uncertainties $(\delta u,\,\delta v)$ can now be calculated by integrating $P^{\text{curt}}_{\delta v}$ over the curtailment region $\mathbb{C}^{\text{curt}}_{(\delta u,\delta v)}$ (Eq. (26)):

$$\delta P^{\text{FBC}}_{(\delta u,\delta v)} \equiv \bar{P^{\text{curt}}_{\delta v}} = -\frac{\Delta t}{\alpha} \int_{-\hat{a}_{\text{sys}}+\delta u}^{\infty} \mathrm{d}\hat{x} \int_{-\hat{x}-2\hat{a}_{\text{sys}}+\delta u+\delta v}^{-\hat{a}_{\text{sys}}+\delta v} \mathrm{d}\hat{y}\,(\hat{y} + \hat{a}_{\text{sys}} - \delta v) f_{XY}(\hat{x},\,\hat{y}) \tag{44}$$

It should be noted that the integration limits for Eqs. (28) and (18) differ, which is due to the fact that the net compliance region $\mathbb{C}^{\text{net}}_{(\delta u,\delta v)}$ and the curtailment region $\mathbb{C}^{\text{curt}}_{(\delta u,\delta v)}$ in the presence of forecast uncertainties are different. Eq. (27) can be re-written as

$$\delta P^{\text{FBC}}_{(\delta u,\delta v)} \equiv \bar{P^{\text{curt}}_{\delta v}} = \chi P_0 \times H(\hat{a}_{\text{sys}} - \delta u,\,\hat{a}_{\text{sys}} - \delta v), \tag{45}$$

with the function $H(c_1, c_2)$ defined earlier (Eq. (14)). It is easy to see that $\partial H/\partial c_1,\, \partial H/\partial c_2 < 0$, so $H(\hat{a}_{\text{sys}} - \delta u,\,\hat{a}_{\text{sys}} - \delta v) > H(\hat{a}_{\text{sys}},\,\hat{a}_{\text{sys}})$ and, consequently, $\delta P^{\text{FBC}}_{(\delta u,\delta v)} > \delta P^{\text{FBC},0}$, i.e. the curtailed power in the presence of forecast uncertainties is higher than in the perfect-forecast case, as expected. Conversely, the net compliance gain is smaller than in the ideal case, as will be discussed in more detail below.

Considering again a bi-variate probability density $f_{\delta U,\delta V}(\delta u,\,\delta v)$ for the forecast errors δu and δv, the expectation value of the curtailment cost in the presence of uncertainties can be written as

$$\langle \delta P^{\text{FBC}}_{(\delta u,\delta v)} \rangle = \chi P_0 \int_0^{2\delta\hat{a}} \mathrm{d}(\delta u) \int_0^{2\delta\hat{a}} \mathrm{d}(\delta v) H(\hat{a}_{\text{sys}} - \delta u,\,\hat{a}_{\text{sys}} - \delta v) f_{\delta U,\delta V}(\delta u,\,\delta v), \tag{46}$$

Assuming for simplicity $f_{\delta U,\delta V}(\delta u,\,\delta v) = \text{const.} = 1/(2\delta\hat{a})^2$ for $\delta u,\,\delta v \in [0,\,2\delta\hat{a}]$ and $f_{\delta U,\delta V}(\delta u,\,\delta v) = 0$ elsewhere, eqn. (45) can be evaluated to yield the following expression:

$$\begin{aligned}\delta P^{\text{FBC}}_{\text{net,ave}}(\hat{a},\,\delta\hat{a}) &= \langle \delta P^{\text{FBC}}_{(\delta u,\delta v)} \rangle \\ &= \chi P_0 S e^{-\hat{a}_{\text{sys}}+\delta\hat{a}} \left\{\frac{1}{2} + e^{-\hat{a}_{\text{sys}}+\delta\hat{a}}\left(-AS - \frac{1}{2}Be^{2\delta\hat{a}}\right)\right\},\end{aligned} \tag{47}$$

where the following two auxiliary functions have been defined:

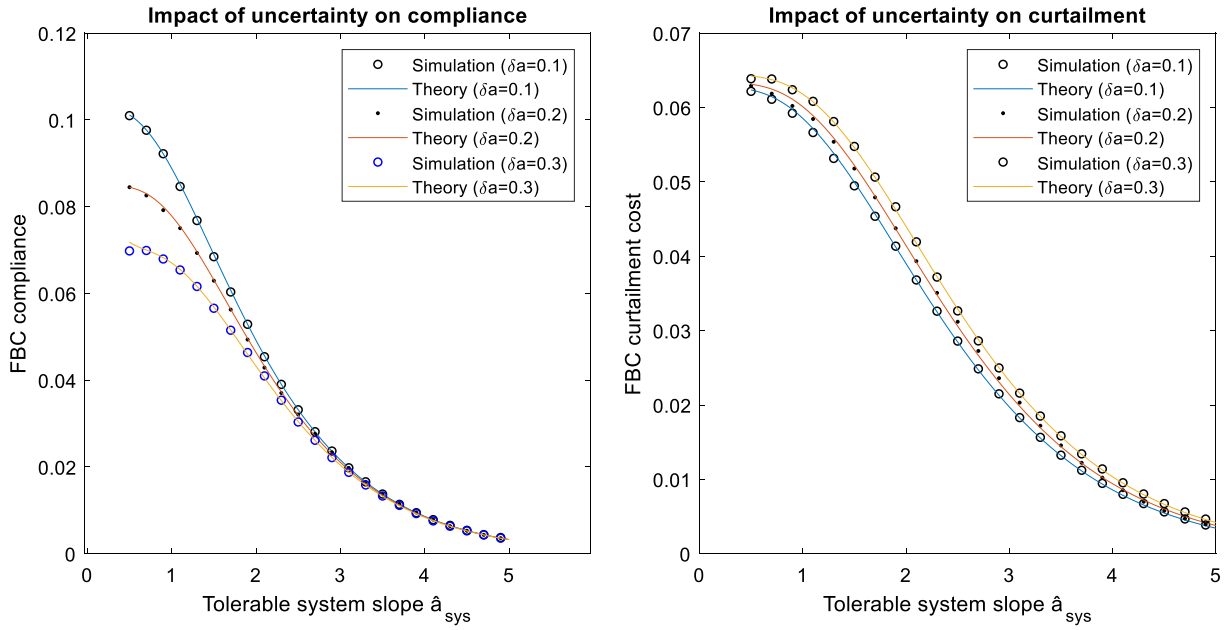


Fig. 7. Impact of forecast uncertainty on compliance (left graph) and curtailment cost (right graph). *Markers*: Simulation results. *Continuous lines*: Analytical results.

$$A(\hat{a}) = \frac{1}{16}(17 + 10\hat{a} + 2\hat{a}^2),\ B(\hat{a}, \delta\hat{a}) = -\frac{1}{4}(2\hat{a} + 5 - 2\delta\hat{a}) \tag{48}$$

$S = S(\delta\hat{a})$ has been defined previously in Eq. (23). It easily verified that Eq. (47) reduces to expression (30) in the case a vanishing forecast uncertainty, i.e. for $\delta\hat{a} \to 0$.

In Fig. 7 the results obtained with the analytical framework for the case of a finite forecast uncertainty have been plotted (continuous lines), together with the corresponding results of the numerical simulation (dots and circles). The analytical and the simulation results can be seen to be almost indistinguishable. As expected, the compliance fraction diminishes with increasing uncertainty, whereas the penalty (i.e. the curtailment cost) increases. It can be seen that at very small normalized tolerable slope values $\hat{a}_{sys}$ the impact on a finite forecast uncertainty is considerable; at more realistic tolerable slope values (say, $\hat{a}_{sys} \geq 3$) the impact becomes rather small. The curtailment penalty, on the other hand, shows little response to forecast uncertainty and remains small in all cases.

4. Validation against empirical data and discussion

After developing the analytical expressions required to quantify the benefits of Forecast-Based Curtailment and its associated (energy) costs, as well as validating those results against simulated data, it is important to address the question of how well the theory describes the real-world operation of a wind farm, given the large number of complexities involved in the latter. Two classes of scrutiny questions can be imagined: (1) how well do real-world data comply with the assumptions of the theory? (2) Does the theory stand up to the results obtained with the empirical data? In other words, can the theory be used for practical design applications? Questions falling into class (1) include the following: (a) do the probability density functions proposed accurately model the empirical data? (b) Is the assumption of uncorrelated subsequent power changes justified? (c) How frequent are outliers (not explicitly modeled in the previous sections), and how do they fit in the picture? Questions in category (2) might address the following issues: (a) does the analytical model provide a good prediction of the empirical compliance fractions? (b) How efficiently are incompliance events mitigated by the FBC approach? (c) Is the effective Laplace model a useful approximation? Though an exhaustive discussion of all the questions above is beyond the scope of the present work, setting part of the agenda for follow-up work, an attempt will be made to address all of the concerns above.

In order to address the research questions posed above, a full year of 1-min power output data from an operational wind farm in Mexico, rated at approximately 50 MW, was analyzed in terms of its suitability for Forecast-Based Curtailment. The data period comprises February 2016 through January 2017. The wind farm data were filtered for a few obvious data handling or transmission glitches, as well as downtimes or missing data periods; otherwise, the data were used as reported. Data recovery after this basic quality assurance was close to 100% in most months, with the exception of October where data recovery was down at 65%. No meta data were made available to the author, but it is reasonable to assume that the wind farm operated uncurtailed for most of the time, given that no transmission restrictions were effective during the period of observations, and system regulation margins were ample. The power output time series of the free-running wind farm were processed numerically through an algorithmic version of the FBC strategy described in Section 3.1. The results from this algorithmic treatment will be referred to as the *empirical FBC (in)compliance ratio* and *empirical FBC efficiency*, respectively. These

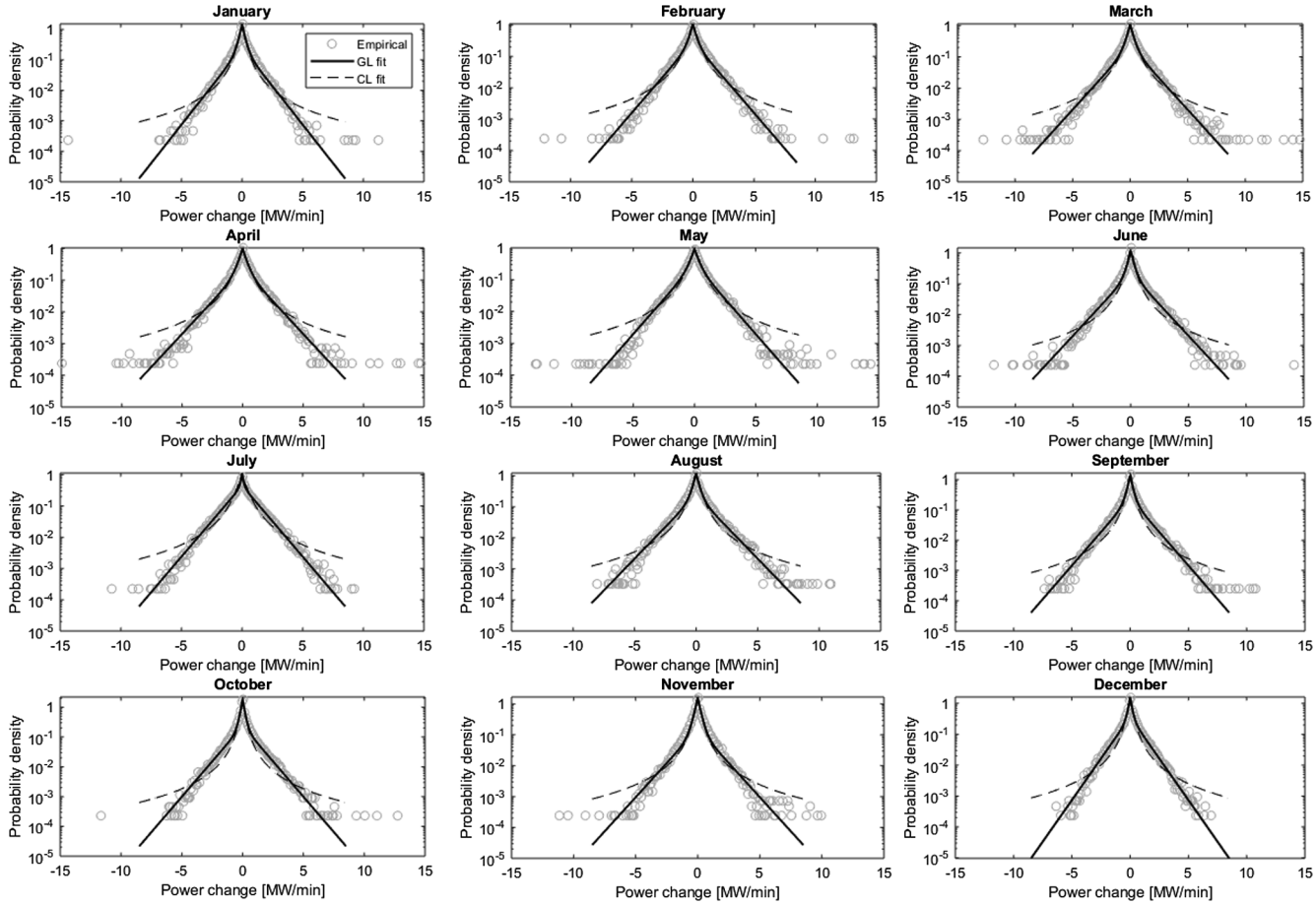


Fig. 8. Empirical and fitted monthly distributions of the power changes of a 50 MW wind farm. *Dots:* Empirical pdf. *GL fit:* Generalized Laplace fit. *CL fit:* Cauchy-Lorentz fit.

data-driven numerical results were then compared against the results of the *analytical* or *theoretical* treatment. The latter requires the use of a probability density function (pdf) of the power changes with its corresponding parameters. This pdf was obtained from the empirical wind farm data through curve fitting. A logarithmic abscissa and custom-written codes were used for this purpose in order to accurately represent data in the long-tail regions of the distribution. Generalized Laplace (GL) distributions provided the best fit to the data, but equivalent (simple) Laplace (L) distributions and Cauchy-Lorentz (CL) were used for reference as well. See Section 2.2 for the corresponding definitions.

4.1. The appropriateness of pdf model

In this section, as in some of the following ones, results are shown for all 12 months in the observation period, in order to allow the reader to reach his or her own conclusions and get some insights into the variability of the empirical data patterns. A natural starting point is the discussion of the empirical probability density functions (pdf) and their relationship to the theoretical models. The results are shown in Fig. 8. Given the importance of the tail regions a logarithmic scale has been used on the vertical axis. It can be seen that the generalized Laplace model (Eq. (2) provides an excellent fit to the data. A large fraction of the data ($=46\% = \langle 1/(1+q)\rangle$) is contained in the fast-decaying part of the distribution, with a characteristic scale of $x_s = 0.2$ MW/min. The remainder of the data can be attributed to the longer tail with an average scale of $\langle x_s/p\rangle = 1.0$ MW/min. The weighted average scale is about 0.66 MW/min. Not unsurprisingly, most power changes X of the wind farm are very small (i.e. $X\Delta t \ll 50$ MW $= P_0$, with $\Delta t = 1$ min as before. As shown in Fig. 8 the range of the X values is generally restricted to the interval [−8 MW/min, +8 MW/min], often even to tighter intervals.

A number of outliers can be observed, though. Negative outliers, the ones potentially relevant to the present study, range between zero (August, September, December), one (January, October) to a few (5–10) for the next of the months. (Note that one event per month corresponds to a pdf-value of 2.3×10^{-4} in the graphs of Fig. 8). Outliers have not been explicitly considered in the theory presented in the present work, and their observed scarcity seems to justify this assumption. Large outliers can of course impact on system operation, since they might surpass a critical value for load shedding, as discussed in Section 2.1, but will be generally irrelevant to the main topic of the work, the mitigation of normal-operation frequency violations. In the case of hypothetical yet realistic 250 MW island grid with a −3 MW/min limit for frequency violations and −15 MW/min limit had been assumed, it can be seen that none of the outliers in Fig. 8 gets anywhere near the load shedding limit, and that the total number of outliers in the [−15 MW/min, −10 MW/min] for the full year amounts to a mere 15. Consequently, a 50 MW wind farm in a 250 MW island grid with the parameters mentioned should not be a cause for load shedding.

4.2. The validity of the uncorrelated consecutive power changes

After checking off two of the items in the first paragraph of Section 4 (appropriateness of the pdf model and outliers) the validity of the assumption of uncorrelated X and Y data will be examined. A natural starting point is a scatter plot and the calculation of the cross correlation coefficient for a given month. As shown in Fig. 9 for March and December, consecutive changes appear to be largely uncorrelated, as substantiated also by the correlation coefficient shown in the graphs. In the case of March the correlation coefficient is very slightly positive (0.03), whereas the correlation for December is slightly negative (−0.12). Negative outliers, taken here as events with $y < -9$MW, and referred to in Section 4.1, are shown for completeness.

Further insights can be gained by calculating the autocorrelation function of the power change time series of each month; see Fig. 10. For March the autocorrelation at the first lag is indeed found to be only very slightly positive, but some longer-range correlation emerges for lags 2–4. This re-emergence of a (small) correlation is indicative of a certain persistence trend which is not modeled in the simplified theory presented here, so some deviation from the ideal results can be expected for March. December shows a smaller positive correlation at lags 2–4, but the overall structure of the autocorrelation function, reminiscent of a strongly damped oscillation, can be seen here as well. Though no fully conclusive argument can be made based on this information it seems clear that the correlation between consecutive changes is small, albeit not zero.

4.3. The accuracy of the theoretical predictions

It is understood that any analytical theory will necessarily have to recur to simplifying assumptions, so the ultimate question is of course to what level of accuracy the relevant results can be predicted. Fig. 11

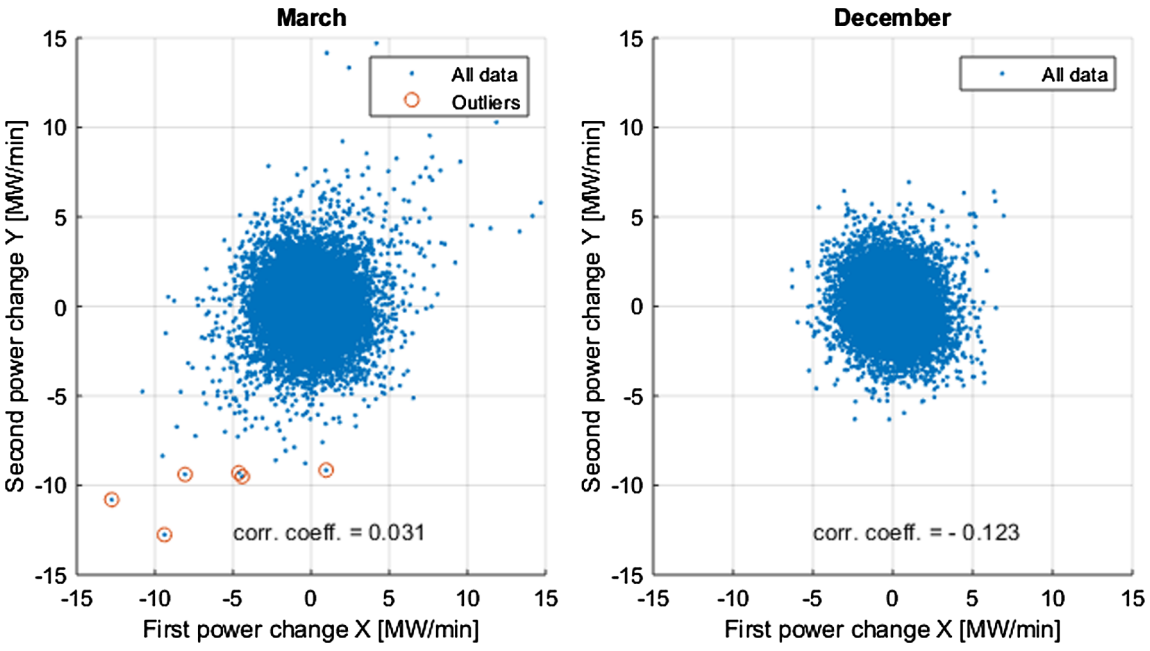


Fig. 9. Scatter plot for consecutive power changes for two representative months.

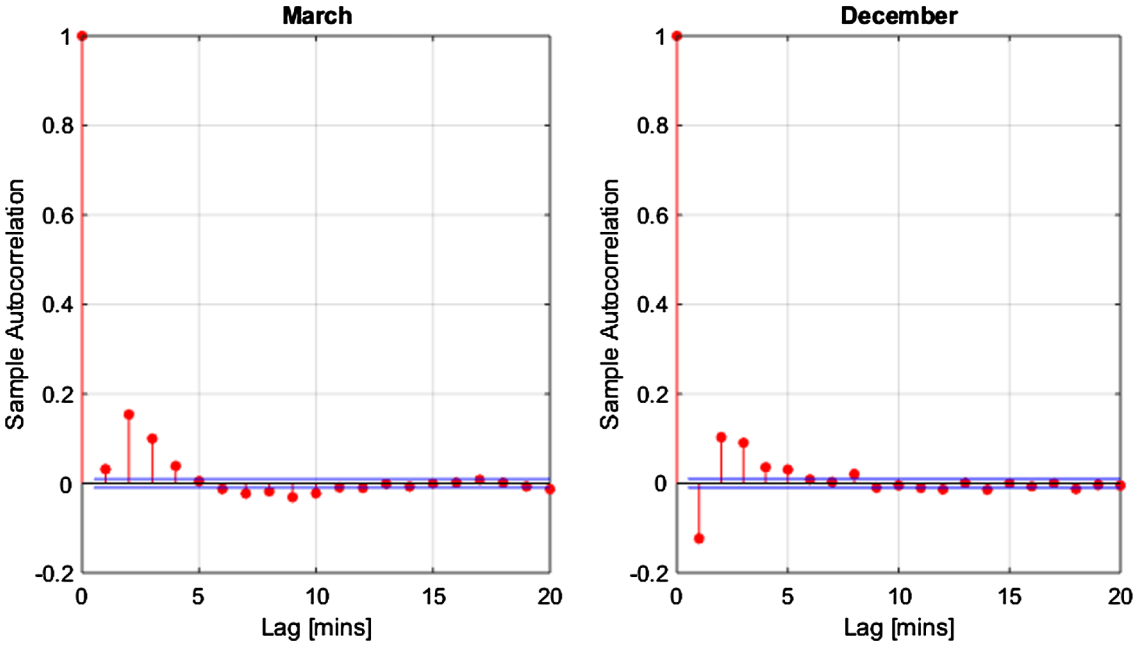


Fig. 10. Autocorrelation function for power changes in two representative months.

shows the overall results of this effort by comparing the total incompliance fraction $1-\rho$ obtained with the FBC-mitigated empirical data set against the predictions of the theory; the generalized Laplace model ($q > 0$) was used. Note that a logarithmic scale has been used on the vertical axis to allow for a distinction between the curves in the tail region, and that the vertical axis spans five orders of magnitude.

It can be seen that the agreement is generally very good up to tolerable slope values of about 3 MW/min; for several months (such as July, September, November, and December) there is even an excellent agreement all through to 5 MW/min. In all other months the net incompliance can be seen to level off between tolerable slope values of 3 and 4 MW/min, resulting in observed incompliance levels higher than the predicted one. Though the difference is clearly conspicuous on the logarithmic scale, it also has to be recognized that the FBC strategy works well for actual wind farm data, and even for the highest permissible slope value of 5 MW/min considered incompliance is low in most months. In most months incompliance with the 5 MW/min limit is observed only a few times a month.

4.4. The effectiveness of the FBC strategy and the appropriateness of the effective Laplace model

In order the quantify the benefit of the FBC approach for mitigation of incompliance events at a given slope level the FBC efficiency (Eq. (12)) is an appropriate measure. In Fig. 12 both the empirical and the predicted efficiencies have been plotted for every month in the observation period. In addition to the full generalized Laplace model efficiencies obtained with the effective Laplace model (Eqs. (19) and (21)) have been added to the graphs. First of all it should be observed that the FBC strategy works generally very well, particularly at high slope values where the support from FBC is critical. Not unexpectedly, after the discussion of the previous results, the months with a good fit of the generalized Laplace model to the pdf have both the best agreements between model predictions and empirical data, and the highest mitigation efficiencies. The model almost perfectly predicts the FBC efficiency curves for July, September, October, and December, and the efficiency in all these cases is close to 100% at a= 5 MW/min; the results are still good for February. In all other months the empirical efficiency does not reach 100% but flattens out around 80% between slope values of about 3 MW/min and 4 MW/min and tends to decline afterwards, although a significant reduction of the incompliance (between 60 and 80%) is observed even at a= 5 MW/min.

The somewhat higher discrepancy in the modeling of FBC efficiency compared to the results in Fig. 11 can be understood if it is observed that an accurate prediction of η_{FBC} requires a high accuracy for the modeling of both ρ_{FBC} and ρ_0, and that η_{FBC} is calculated as the quotient of two small numbers. Considering this challenge the accuracy is remarkably high. As a footnote it should be remarked that ρ_{FBC} depends on an accurate model of the bi-variate pdf, whereas ρ_0 depends on the univariate distribution, so the deviation of the real distribution of data from the model affects both terms in different ways.

As far as the effective Laplace model is concerned, it can be seen from Fig. 12 that this model generally tends to overpredict the empirical FBC efficiency but the discrepancy is not very large. It should be noted that no adjustment to the fit parameters has been made, and that the effective fluctuation scale has been obtained directly from Eq. (21) in all cases. Therefore, the model may be well suited for an initial estimation of the potential for FBC in a given site, given that the formulas can be easily implemented in a spreadsheet application.

Returning to the subject of wind farm sizing introduced in Section 3.2, a few rough calculations can be made at this point. The natural incompliance fraction $1-r_0$ of the operating wind farm at a critical slope of $a = 3$ MW/min averages about 10^{-2}, corresponding to some 3 incompliance events per day. The FBC-enhanced incompliance fraction $1-r_0-r^{\mathrm{FBC}}$, as shown in Fig. 11, varies between about 10^{-4} for December and 2×10^{-3} for April, with an average of 1.6×10^{-3}. This correspond to a six-fold reduction in incompliance or an average mitigation efficiency of 84%, in rough consistency with Fig. 12. Assuming for simplicity that the incompliance fraction at $a = 3$ MW/min acceptable to the grid operator is 10^{-2}, then the P_0= 50 MW wind farm would just have the maximal acceptable size if operating under natural compliance conditions. If equipped with an FBC control the incompliance rate is reduced by a factor of 6, so the allowed wind farm size can be larger. Solving eqn. (31) for the admissible scale x_s and using $x_s = \chi P_0$ the new admissible wind farm size $P_{0,\mathrm{new}}$ can be calculated. Using the analytical expressions of Eq. (20) a value of $P_{0,\mathrm{new}} = 90$ MW is obtained, i.e. an 80% increase in name-plate capacity. This theoretical result was found to be consistent with a compliance simulation based on the empirical data, where the wind farm size was increased to 90 MW. Though these back-of-the-envelope calculations

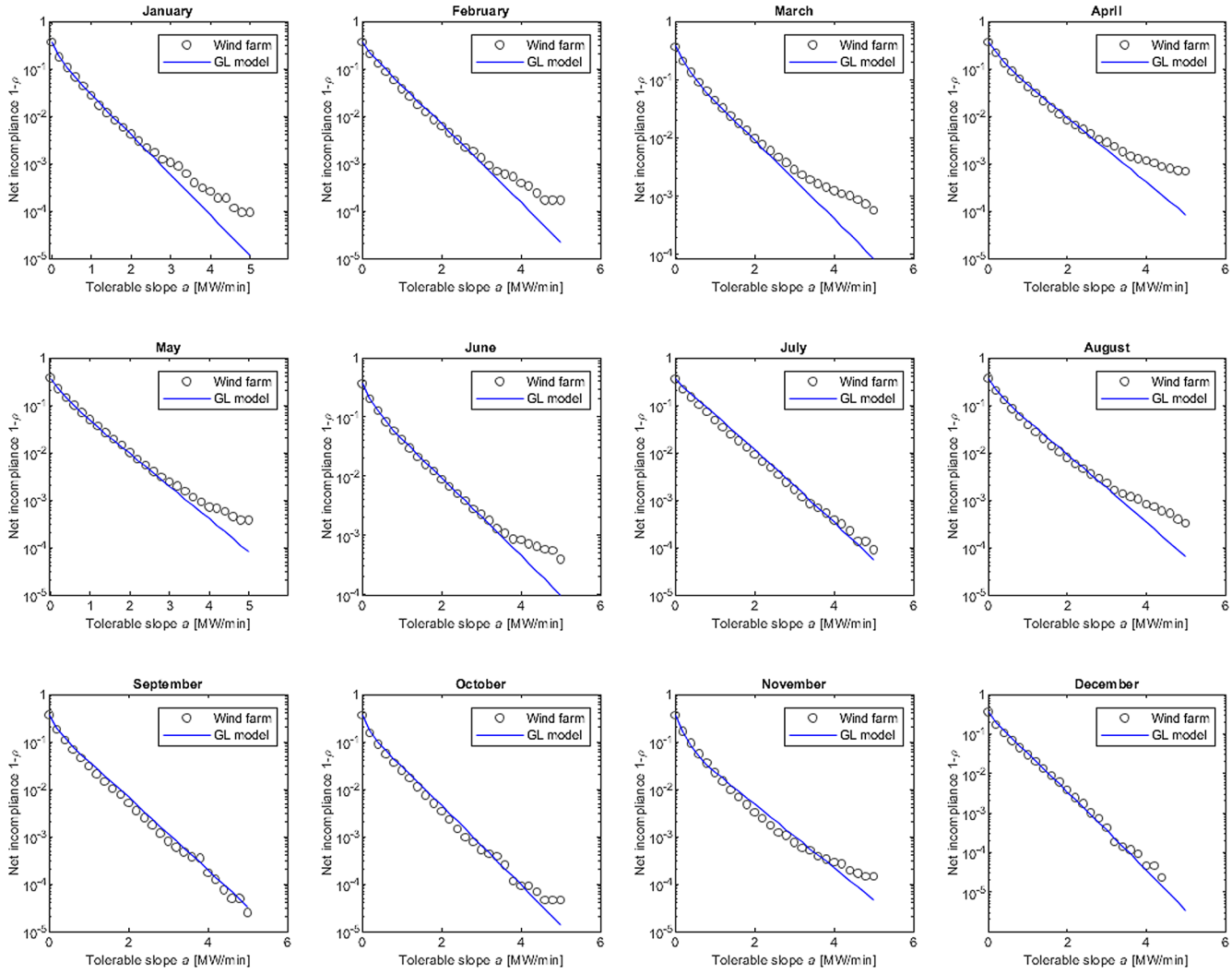


Fig. 11. Incompliance fractions calculated from data of an operating wind farm rated at 50 MW, assumed to be operating with the FBC strategy. *Circles:* FBC incompliance fractions calculated numerically from wind farm time series. *Continuous lines*: Predictions of the analytical model based on a generalized Laplace fit to the empirical pdf.

have to be taken with a grain of salt they do illustrate the effectiveness of the FBC approach, as well as the usefulness of the analytical framework.

As a final point it should be mentioned that the indicative calculations presented above correspond to perfect forecasts ($\delta a = 0$). The argumentation can be extended, however, to include a finite forecast accuracy. Eq. (24) has then to be replaced by $\rho_0(a_{\text{crit}}/x_{s,\text{eff}}) + \rho^{\text{FBC}}(a_{\text{crit}}/x_{s,\text{eff}}, \delta a/x_{s,\text{eff}}) = \rho_{\text{req}}$, where it should be noted that δa at a given site can be expected to be a decreasing function of the installed wind farm capacity ($\partial \delta a/\partial P_0 < 0$), given that the output of larger wind farms is easier to forecast.

4.5. Can the theory be extended to include long-term ramps?

Wind farms are known to show long ramps, i.e. periods of time where the output power consistently rises or falls over extended periods of time, typically ranging from few tens of minutes to several hours. Given that these periods are typically long on the 1-min scale the bivariate 1-min distribution can be modeled to a good approximation as $f_{XY}(x-s, y-s)$, where s is the average slope of the ramp. Empirical bivariate slope distributions $g_{SL}(s, l)$ can be constructed from the data (with l being the length of the ramp), from which a marginal distribution $g_S(s)$ can be calculated. The theory in Section 3 can then be easily generalized to the case of ramps. With the empirical data set of Section 4 it was found that the net effect of ramps was very small; it should be pointed out, though, that the identification of (long) ramps is not trivial, and that the specific algorithm used for their identification may have an impact on the distribution of the short-term fluctuations discussed in this paper; see, e.g., Quyang et al. [16] for a discussion of this coupling, albeit in a different context. Given the limited space it will not be elaborated on this subject. An in-depth discussion will be provided in follow-up work.

5. Summary and conclusions

In the present work, a novel approach to wind farm curtailment for short-term slope control has been proposed, based on the assumption that future normal-operation power output values can be predicted relatively accurately, typically two minutes into the future. This new strategy has been termed *Forecast-Based Curtailment* or FBC. The paper provides a comprehensive analytical framework for the understanding and calculation of the fractions of time during which a wind farm *naturally* complies with a given slope limit (ρ_0) and the additional compliance fraction (ρ_{FBC}) which can be obtained through the FBC method proposed. Expressions for the cost of FBC-curtailment have also been derived. In general, the FBC method seems to be a highly cost-effective strategy to reduce fluctuations, given both the high level of

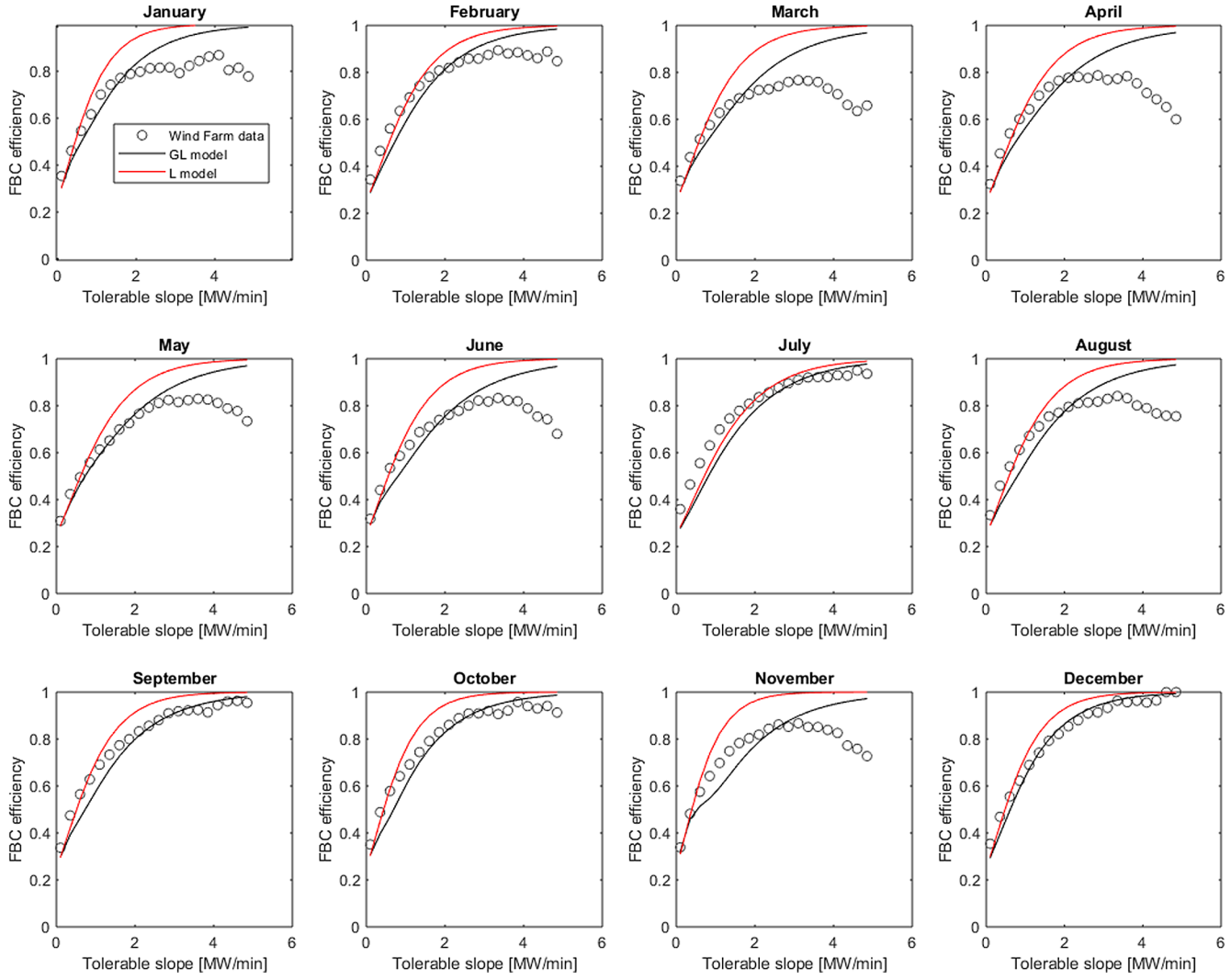


Fig. 12. FBC efficiency calculated from data of an operating wind farm. *Circles:* FBC efficiency values calculated numerically from wind farm time series. *Black lines*: predictions of the analytical model based on a generalized Laplace fit to the empirical pdf. *Red lines*: analytical predictions of the equivalent simple Laplace model. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

mitigation of incompliance events (as measured by the *FBC efficiency*) and the very low level of power not delivered. Ideal FBC efficiency values tend to 100% for large values of the slope limit, and even with actual wind farm data the efficiency is generally consistently as high as 80% (and often even near the theoretical limit). While the FBC results already speak for themselves the cost-benefit ratio becomes particularly impressive if compared to a standard strategy of slope control, *flat curtailment*, where the wind farm output is de-rated at a constant factor at all times. More sophisticated strategies are of course possible, but given its role as a reference strategy the simple de-rating approach was deemed good enough. A detailed comparison of different curtailment strategies is of course an interesting undertaking but is outside the scope of the present work. Combinations of FBC with flat curtailment are possible and maybe an appropriate choice if slope compliance requirements are very stringent. Interestingly, flat curtailment also boosts the effect of FBC by reducing the effective scale of the fluctuations.

An important question is of course how forecast uncertainties affect the FBC compliance fraction and the corresponding curtailment cost. Analytical expressions have also been calculated for this more general case, and the results have been validated against numerical simulations based on synthetic data, finding an excellent agreement. The subject of how well the wind farm output up to two minutes into the future can be predicted is outside the scope of the present work, although it was suggested that a combination of stochastic modeling with upstream measurements might be accurate enough for the present purposes. Detailed analyses will be presented in follow-up work.

Though the larger part of this paper was dedicated to the analytic framework of the FBC strategy and its validation against simulated results based on synthetic data an assessment of a one-year data set of 1-minute records of an operating wind farm was also included. The operational data were first used to validate the basic assumptions of the FBC theory, such as negligible correlation between consecutive 1-min power changes and a good fit to a generalized Laplace probability density function. A numerical version of the FBC strategy was then implemented on 12 monthly data sets, in order to assess both the overall performance and to detect variations. An excellent agreement of the total (FBC-mitigated) compliance with the theoretical predictions was found for critical slope values of up to about 5 times the characteristic fluctuation scale $\langle x_{s,\mathrm{eff}} \rangle$, and still a fair prediction up to 7.5 $\langle x_{s,\mathrm{eff}} \rangle$. Based on the results presented it is believed that this work will help mitigate the concerns of grid operators and utilities regarding wind farm-induced concerns about frequency regulation and promote low-cost solutions for the massive grid integration of wind energy required to meet climate change mitigation goals.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

Support from the Energy and Climate Change group at Tecnológico de Monterrey is gratefully acknowledged. I would also like to thank Julio Gutiérrez for proofreading and useful suggestions.

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